Year II — Algebra

Based on lectures by Various Lecturers

Notes taken by Anonymous

Michaelmas - Epiphany 2026-2027

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0.1 Curves 1

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Definition 0.1.1 (Curve). A curve C is a 1-d subset of $\mathbb{R}^n$, for $n > 1$. It is usually convenient to describe a curve by a parametrisation, $\boldsymbol{x}(t)$, which maps a real interval $[t_0, t_1]$ into $\mathbb{R}^n$. This is a vector-valued function

$$\mathbf{x}(t) = x_1(t)\mathbf{e}_2 + x_2(t)\mathbf{e}_2 + \ldots + x_n(t)\mathbf{e}_n,$$

where $\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\}$ denote the Cartesian unit vectors

Theorem 0.1.2 (Curve). A curve C is a 1-d subset of $\mathbb{R}^n$, for $n > 1$. It is usually convenient to describe a curve by a parametrisation, $\boldsymbol{x}(t)$, which maps a real interval $[t_0, t_1]$ into $\mathbb{R}^n$. This is a vector-valued function

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Result 0.1.3 (Curve). A curve C is a 1-d subset of $\mathbb{R}^n$, for $n > 1$. It is usually convenient to describe a curve by a parametrisation, $\boldsymbol{x}(t)$, which maps a real interval $[t_0, t_1]$ into $\mathbb{R}^n$. This is a vector-valued function

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Corollary 0.1.4 (Curve). A curve C is a 1-d subset of $\mathbb{R}^n$, for $n > 1$. It is usually convenient to describe a curve by a parametrisation, $\boldsymbol{x}(t)$, which maps a real interval $[t_0, t_1]$ into $\mathbb{R}^n$. This is a vector-valued function

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Lemma 0.1.5 (Curve). A curve C is a 1-d subset of $\mathbb{R}^n$, for $n > 1$. It is usually convenient to describe a curve by a parametrisation, $\boldsymbol{x}(t)$, which maps a real interval $[t_0, t_1]$ into $\mathbb{R}^n$. This is a vector-valued function

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Question 1

Let $X \sim N(20, 3^2)$. Find $P(X > 25)$ and anything else you want.

Proof of 'Theorem 3'. Here we have □

Solution to 'Question 3'. Here we have

Suppose we were to now try working this out

$$xy^2$$